

Jurassic Spark: Analog Computer

ECEN 4620

Control theory and feedback loops are fundamental aspects of electrical engineering. However, students often struggle to understand these concepts due to abstract control equations and simulations that hide real-world effects. Our Analog Computer addresses this with swappable analog modules, a mechatronic ball-levitation demonstrative system, and an embedded oscilloscope — giving students a fully tactile, visual experience with closed-loop control.

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1 Executive Summary

1.1 Product Overview

High-Level Operation

First, based on pre-determined control logic, the student will insert some assortment of the four modules (i.e., integrator, gain, subtractor, and adder) into the computer housing.

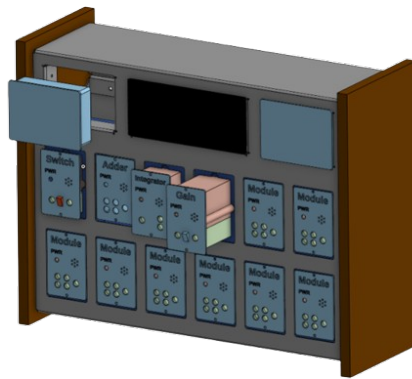


Figure 1: Modules being inserted into computer chassis.

From here, the student will begin developing the control signal. The next step is to route the output of the sensor from the mechatronic system to the computer, which will provide information about where the ball is positioned in the tube.

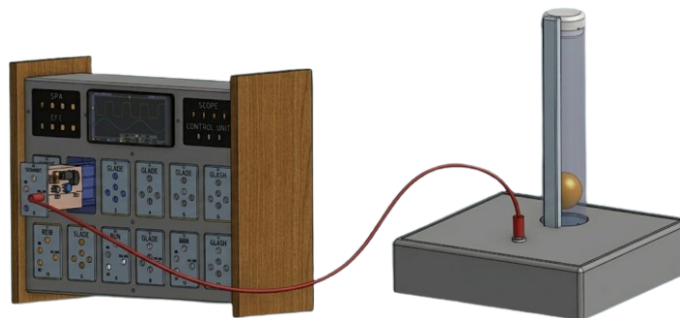


Figure 2: Information about the current ball height comes from the sensor in the mechatronic system.

To establish a reference voltage, the student must connect the 12V supply located at the top of the computer to a gain module set in the 0-1 mode. Continuing to connect the inputs and outputs between the modules, the student develops a control signal. This control signal feeds back into the mechatronic system, closing the loop.

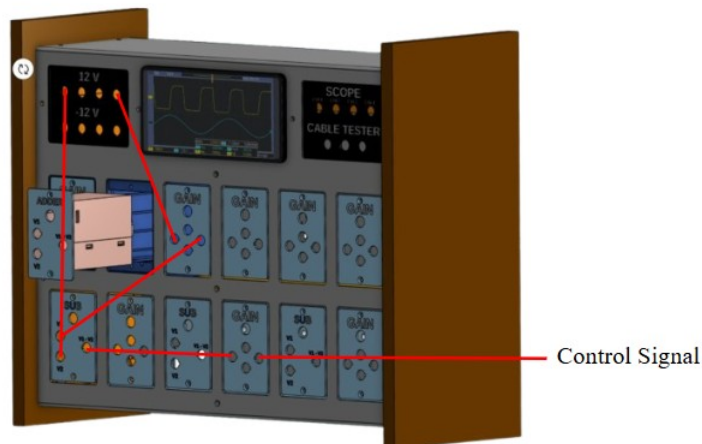


Figure 3: Routing the inputs and outputs of the modules develops a control signal.

Finally, with the control signal established and the feedback loop closed, the student observes the ball maintain a constant height as the fan in the mechatronic system dynamically adjusts its speed. Moreover, the student can compare the reference voltage with the voltage corresponding to the height of the ball, for example, by connecting these signals to the embedded oscilloscope on the computer.

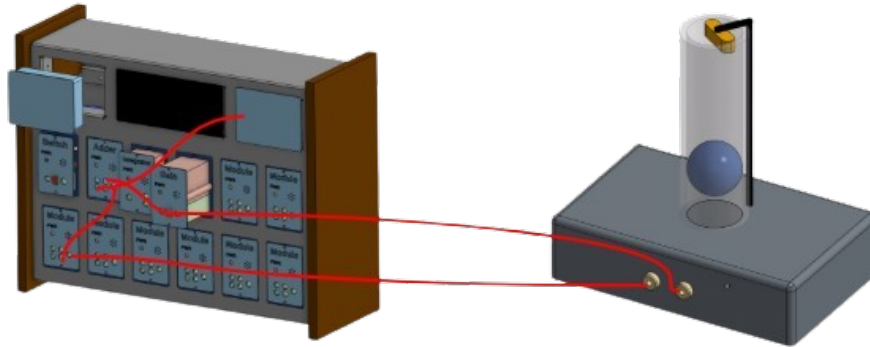


Figure 4: Full operation of analog computer.

1.2 Value

The underlying objective of this product is to create an interactive and intuitive platform for students to experiment with control laws. One of the key ways this is accomplished is by creating a design in which multiple different feedback operations can be implemented and tested quickly. For example, the figure below shows the feedback path we demonstrated at EXPO.

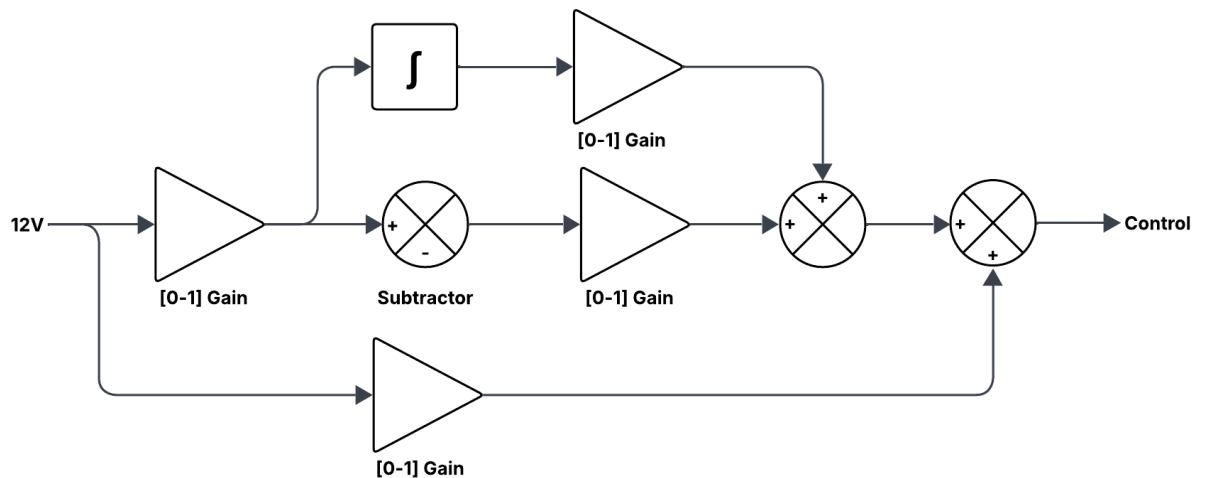


Figure 5: Feedback path demonstrated at EXPO.

This feedback configuration represents only one example of the many control laws that can be implemented using the Analog Computer. By rearranging modules and modifying signal paths, students can create a wide range of feedback architectures to explore different system responses and controller behaviors. The modular design encourages experimentation by allowing users to quickly insert, remove, or reorder operations without rebuilding an entire circuit. Each gain module also includes a selectable range switch and potentiometer, enabling users to tune the gain continuously within either a 0–1 range or a 1–10 range. This flexibility allows for both fine adjustments and larger scaling changes, giving students greater control when tuning feedback performance.

Another key source of value comes from the system’s ability to bridge the gap between theoretical control concepts and physical behavior. By allowing students to directly observe how changes in feedback structure, gain, and signal flow affect a real system, the Analog Computer creates a more intuitive learning experience than simulations alone. Students can quickly compare different controller configurations, visualize signal responses in real time, and develop a stronger understanding of how control laws influence system stability and performance. The modular and durable design also supports repeated experimentation, making the platform well suited for long-term educational use in laboratory and classroom environments.

1.3 Key Features

Several key features were incorporated into the Analog Computer to improve usability, safety, and educational value. Each module includes an LED saturation indicator that alerts the user when the internal op-amp is approaching or reaching its output limits, helping students recognize nonlinear behavior and understand the practical limitations of analog control systems.

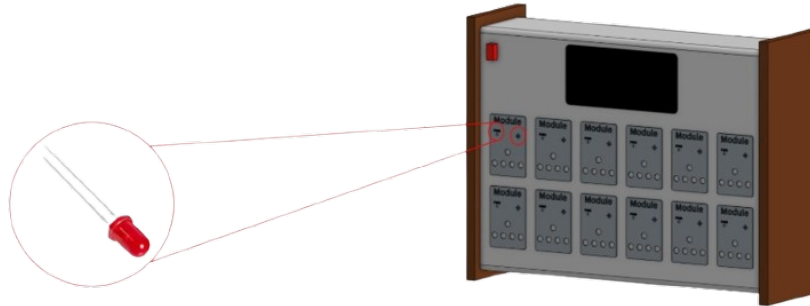


Figure 6: LED on each module conveys saturation.

Additionally, custom input and output connectors are also used throughout the platform to reduce the possibility of miswiring, ensuring that signals are routed correctly between modules and protecting the system from incorrect connections.

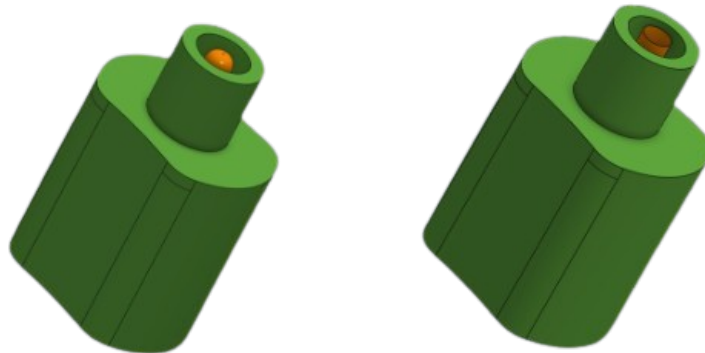


Figure 7

Finally, the modular architecture allows students to quickly insert, remove, and rearrange functional blocks without rewiring an entire circuit, enabling rapid experimentation with different feedback paths. Integrated signal monitoring further enhances the learning experience by allowing users to observe control signals and intermediate node voltages in real time, providing immediate visual feedback during system tuning and analysis.

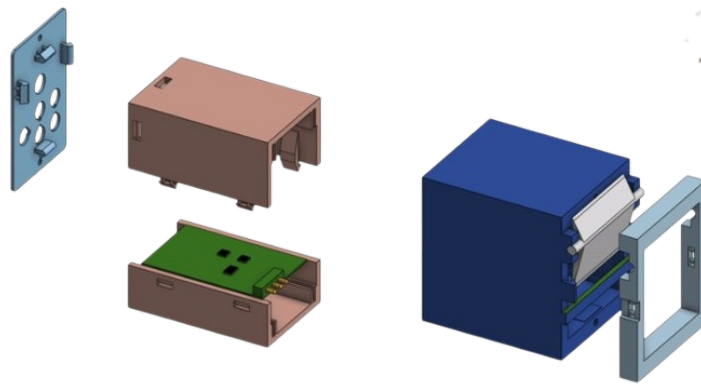


Figure 8: Housing for PCB modules.

1.4 Performance

Overall, the Analog Computer performed well and successfully met the primary requirements established for the intended user experience. The system allowed students to construct and modify feedback paths using interchangeable modules, observe system response in real time, and interact with a physical mechatronic demonstration. During testing and demonstration, the platform was capable of producing stable control behavior and provided a clear visualization of how different control laws influenced system performance. The modular design also functioned as intended, allowing modules to be inserted, removed, and rearranged quickly without requiring extensive setup time.

While the system achieved its fundamental objectives, there remains significant opportunity for refinement and improvement. One of the most noticeable limitations was voltage noise present within the integrated oscilloscope display, which occasionally reduced signal clarity and made small waveform details more difficult to interpret. Additional filtering, improved grounding strategies, and further PCB layout optimization could reduce this noise and improve measurement quality. Future iterations could also improve overall packaging, signal routing, and subsystem integration to create a more polished and robust educational platform.

2 Requirements

The requirements for the Analog Computer were developed to ensure the system provided an intuitive, safe, and educational platform for experimenting with control theory concepts. These requirements originated from both customer expectations and engineering constraints, focusing on modularity, ease of use, signal visualization, and physical demonstration of feedback control. The intended customer experience centered around allowing students to quickly assemble feedback paths, observe real-time system response, and interact with a tangible control system rather than relying solely on simulation software.

Several primary use cases guided the design of the product:

- Students select and insert modules corresponding to desired control operations such as gain, integration, addition, or subtraction.
- Students route sensor signals between modules to construct a feedback path and generate a control signal.
- Students adjust potentiometers on gain modules to tune controller response and observe system behavior.
- Students connect signals to the integrated display to visualize waveforms and intermediate control signals in real time.
- Students close the loop with the mechatronic system and observe the ping pong ball maintain a stable height within the tube.
- Professors can rearrange modules quickly to demonstrate multiple control laws during classroom instruction.

Several marketing and engineering requirements were successfully implemented in the final design:

- Modular add/subtract, gain, integrator, and switch functionality.
- Swappable modules capable of being inserted and removed quickly.
- Integrated oscilloscope display for real-time signal monitoring.
- Mechatronic feedback demonstration using a fan-driven ping pong ball system.

- Saturation indication using LEDs to alert users when op-amp limits are reached.
- Custom connector system designed to reduce miswiring between inputs and outputs.
- Compact, bench-top enclosure suitable for classroom and laboratory use.
- Fully manufacturable system using on-campus fabrication resources.

While many core requirements were achieved, several features remained stretch goals or areas for future refinement:

- Improved oscilloscope signal quality with reduced electrical noise.
- Additional module types such as comparator, inverter, or low-pass filter modules.
- Expanded tertiary systems beyond the ping pong ball demonstration.
- More advanced data logging and waveform storage functionality.
- Increased robustness of signal routing and cable management.

3 The Technology Options

3.1 Analog Computer

The analog computer, being the main component of our project, deserved some of the most consideration for technology options. Due to our requirements aforementioned, we knew we wanted a robust and high-performing RRIO op-amp, and slew rate wasn't that big of a concern, since these control loops are going to be "slow" dc-circuits. With these requirements, we considered using an op-amp that every electrical engineering student at CU Boulder is familiar with - the TLV272 from Texas Instruments. Ultimately, this didn't meet our desire to have a "12 volt bipolar" system, as the TLV272 can only operate on up to 16 volts. However, this provided a blueprint to find the op-amp we ultimately went with - Texas Instrument's OPA196 family. With it being operable to up to 32 volts, as well as some quality of life features baked in such as overvoltage and short circuit/overcurrent protections, it made this the clear choice to help meet our requirement of a robust and "student-proof" system as well, and has proven to be a very effective and accurate op-amp for our system.

To meet the requirements of detecting saturation, we chose to go with a dual package LM393 comparator to build a window-comparator circuit to detect when our op-amp outputs are swinging close to the max output (with a .5 volt window from either power rail). The LM393 was a clear choice for this application, as really just about any comparator that accepts our voltage range of 0-24 volts would work, and the LM393 in particular is a tried-and-true component, being extremely cheap and extremely common.

3.2 Oscilloscope

The embedded oscilloscope was the "subsection" of our technology requirements in which we had possibly the most freedom, as there were not a lot of strict requirements or requirements at all for how to implement this feature. Ultimately, we knew that it didn't have to be the most precise or high-speed oscilloscope, and could be a rather "bad" oscilloscope by modern standards, as it will be used primarily to look at DC signals and nothing higher than 100 Hz, if at all. Thus, there was a lot of avenues we could have taken to meet those basic requirements, even going as far as to implement a complete system such as the Analog Discovery 2 from Digilent, which would have the benefit of being way more accurate and better made than anything we'd actually need in this project, as well as a

strong likelihood that the students taking this course would already be in part familiar with this system as they use this device in the electronics design laboratory course, a prerequisite for electrical engineering students.

However, because our sort of atypical system being a pseudo-bipolar system, we ultimately decided on doing a custom solution. For a custom solution that requires a front-end, a Raspberry Pi is the clear choice, due to its ability to give us access to all the power and flexibility with Linux. Out of the Raspberry Pi family of micro controllers, we decided on using a Zero 2 W, primarily for physical form-factor concerns, while still having the ability to use wireless features including WiFi and Bluetooth. Ultimately, we ended up using Qt as our application development framework, as it's very robust and has great documentation, as well as literally had an oscilloscope example baked in to its IDE, and greatly simplified the front-end development cycle. C++ was incorporated for data processing and communication to our ADC.

As far as the ADC we chose, we ended up going with a Waveshare Raspberry Pi hat breakout module built around the ADS1263, a 10-channel 32 kSPS ADC. This means that with 4 channels integrated, we were getting around 8 kSPS per channel, which was more than fine for the scope of this project. This required a secondary custom PCB to level shift inputs down to the acceptable 5 volt range, however, which was not ideal compared to our original custom PCB hat solution that would do that along with powering the Pi itself.

3.3 Mechatronic System

The primary concern for the mechatronic system ended up being the sensor, and how we sense the ball overall. Multiple sensors and methodologies were attempted, including ultrasonic sensors, an analog time-of-flight sensor, and multiple digital time-of-flight sensors. The primary issue was the ultrasonic sensor had slight issues with the wind from the fan affecting readings (not to mention not being too accurate in the first place), and a combination of issues with light-based sensing methods, primarily due to the light being internally reflected in the acrylic tube, which was a strict requirement in this project, so students could actually see the ball inside our tube. Due to the reflections we were experiencing, we looked into sensors that had the ability to adjust its region of interest for measurements, which led us to a breakout module from Adafruit that incorporates ST's VL53L1x time of flight sensor. It has a 4 m range, and allows you to adjust the region of interest down to a 5 degree "cone", which proved to be absolutely vital for the success of this system and was implemented with great success.

The fan was a driving factor for a lot of our choices overall for mechatronic system, as we wanted to pick something that would be relatively easy to implement, rather than having to deal with motor control issues ourselves, while still meeting our requirements of "analog control", and being powerful enough to actually lift the ball with high static-pressure ratings. Because of these considerations, we decided to go with a PC case fan, as it would be easily controllable with a PWM signal and greatly simplify aspects of our integration. The specific fan we chose is the NF-F12 industrialPPC-3000 PWM from Noctua, as it was designed as a server-rack fan and thus has high static pressure ratings (to get good airflow in constricted environments), while still having all the features and ease of use of a traditional PC fan.

To interface these components and act as a sort of "translation layer" between everything, we needed a microcontroller. We ultimately decided to go with an Arduino Nano, as they are extremely simple to work with, and had libraries readily available for the sensor we were using. It would effectively read the signal passed from the computer, map that directly to a PWM duty-cycle for the fan, and spit the sensor reading values out as analog voltages to be sent back to the Analog Computer.

4 The How

4.1 Functional Block Diagram

The Analog Computer can be understood as a closed-loop system that connects four major subsystems: the **Bank of Modules**, the **Integrated Oscilloscope System**, the **Power Distribution Unit (PDU)**, and the **Mechatronic Set-Up**.

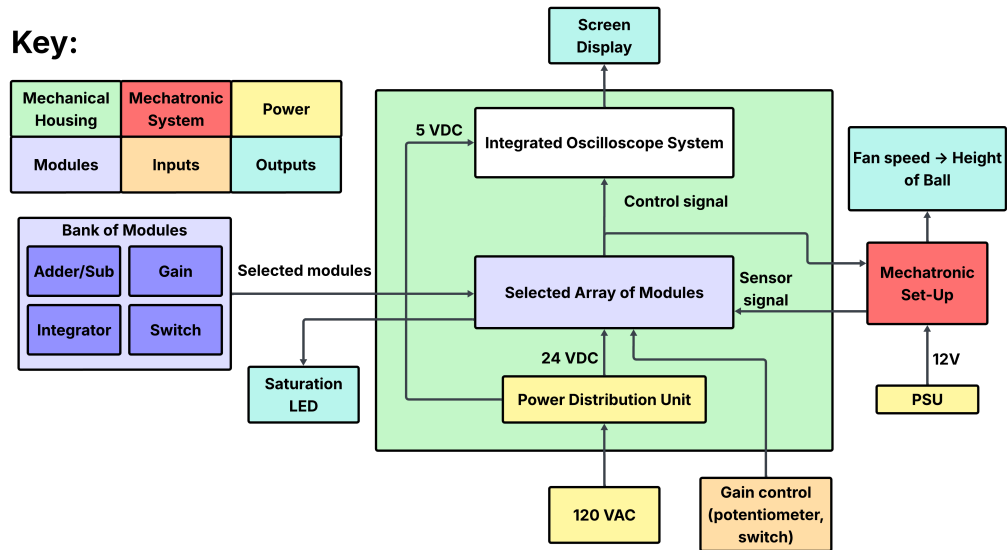


Figure 9: Functional block diagram for the Analog Computer.

4.2 Bank of Modules

4.2.1 Module Mechanical Design

Each module consists of a 3D-printed enclosure that houses a custom PCB. Power is delivered to each module passively via pogo pins on the rear face that mate with a spring-contact rail on the back-plane when the module is inserted, eliminating the need for a separate power connector. Banana jacks on the front face expose the signal input and output ports. A front-mounted indicator LED and buzzer driven by an LM393-based saturation detection circuit alert the student when the op-amp output approaches the supply rails.

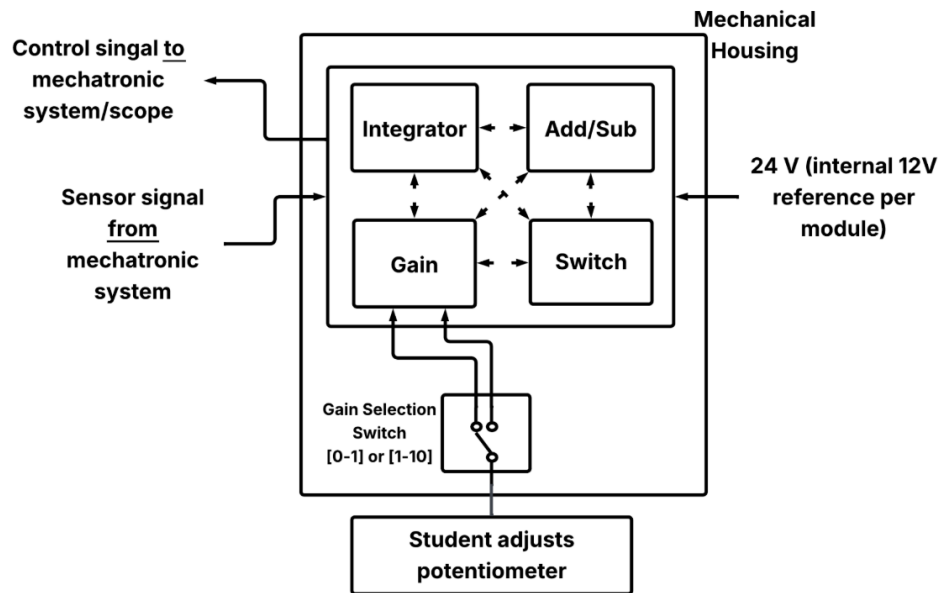


Figure 10: Functional Block Diagram of the Modules

4.2.2 Adder / Subtractor Module

The adder/subtractor is a single PCB implementing a classic differential-amplifier topology using OPA196 op-amps. Two banana-jack inputs feed the non-inverting (add) path and two feed the inverting (subtract) path through matched 100 kΩ resistors, giving the output:

$$V_{\text{add}} = 12 - (V_{\text{in1}} + V_{\text{in2}})$$

A 12 V virtual-ground reference generated on-module from the 24 V supply defines the signal midpoint, so the usable output swing spans −12 V to +12 V. The module does not include saturation detection, because the differential topology self-limits at the supply rails without runaway behaviour.

4.2.3 Gain Module

The gain module provides selectable gain ranges of [0, 1] and [0, 10] via a front-panel toggle switch. Each range is implemented with a separate inverting amplifier stage using OPA2196 op-amps, and an additional inverter stage corrects the sign inversion introduced by the amplifier so the output polarity matches the input. A

100 k Ω potentiometer accessible on the front panel sets the feedback resistance, allowing the student to tune the gain continuously within the selected range. An LM393 window comparator monitors the output and illuminates the saturation LED/buzzer when $|V_{\text{out}}|$ approaches 12 V.

4.2.4 Integrator Module

The integrator module uses an OPA2196 in the classic integrator configuration. The 12 V virtual-ground reference keeps the quiescent output centered at mid-rail. The same LM393-based saturation sensing used in the gain module is included here because the integrator is the module most susceptible to slow drift to saturation under student use. Integration is susceptible to continued integration on small DC offsets, requiring the saturation detection to be present so students will know when that drift has occurred over time.

4.2.5 Switch Module

The switch module routes one of two selectable input signals to a single output. Students use it to inject a reference setpoint into the control loop or to toggle between two signal paths mid-experiment. A front-panel toggle switch physically selects between the two banana-jack inputs; no active circuitry is required, keeping the module simple and reliable.

4.3 Power Distribution Unit

The PDU accepts 120 VAC from a standard three-prong wall outlet. A Mean Well LRS-150-24 enclosed switching power supply converts the mains input to a regulated 24 VDC bus at up to 6.25 A (150 W). The 24 V bus is distributed to each module slot via a custom PCB that provides one pogo-pin rail contact per slot; adding or removing modules changes the load but does not interrupt power to the remaining slots. A Traco Power TSR 2-2450N non-isolated buck converter steps the 24 V rail down to a stable 5 V output to power the Raspberry Pi Zero 2W and the LCD screen.

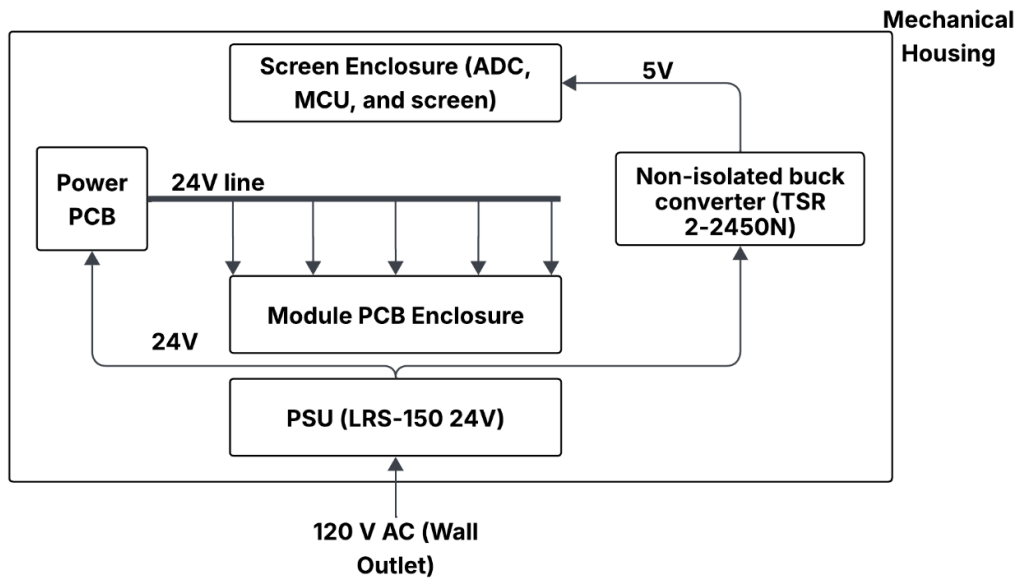


Figure 11: Functional Block Diagram of the Power System

4.4 Integrated Oscilloscope System

This system replaces an external bench oscilloscope with a self-contained, embedded display that students could use to observe control signals in real time without additional equipment.

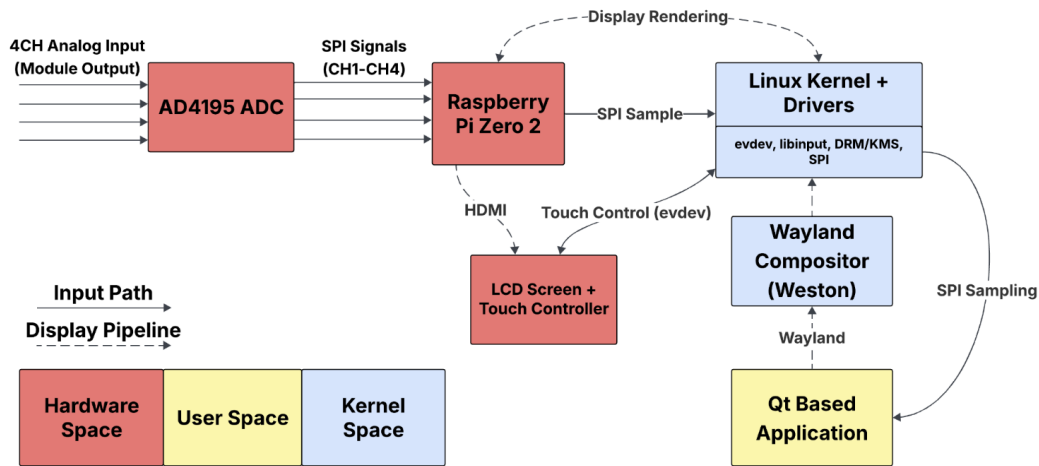


Figure 12: Functional Block Diagram of the Screen System

4.5 Mechatronic Set-Up

The mechatronic system is a self-contained ball-levitation demonstration unit powered separately from a 12 VDC supply. It serves as the real-world plant that students control using the module array.

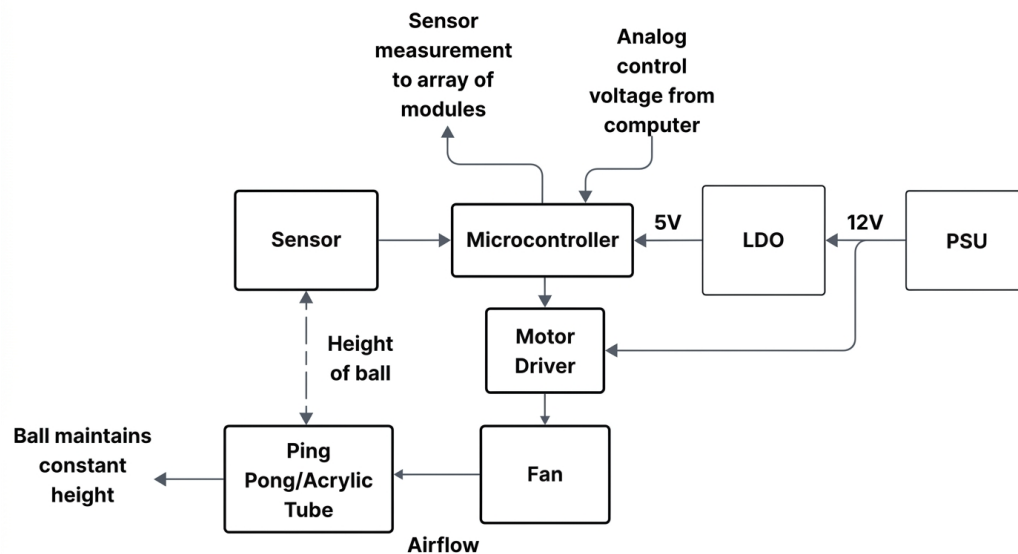


Figure 13: Functional Block Diagram of the Mechatronic Set-up

5 Analysis

5.1 System-Level Performance

The Analog Computer successfully met its primary objective of providing a functional, hands-on platform for closed-loop control experimentation. During EXPO demonstration, the system achieved stable ball levitation within the mechatronic tube, with the fan responding dynamically to the control signal generated by the module array. People were able to construct a feedback path, observe the system response in real time on the embedded oscilloscope, and adjust gain to modify closed-loop behavior thus validating the core use case the system was designed for.

Of the marketing and engineering requirements established at the outset of the project, the majority were successfully implemented in the final design. Modular add/subtract, gain, integrator, and switch functionality was fully implemented. Modules can be inserted and removed from the backplane in seconds, and the pogo-pin power delivery system eliminated the need for any separate power connections when swapping modules. The custom connector system reduced miswiring by ensuring inputs and outputs could not be accidentally swapped. The platform was compact enough to be operated on a standard laboratory bench and was fabricated entirely using on-campus resources.

5.2 Oscilloscope and ADC Performance

The embedded oscilloscope system was built around the ADS1263 32-bit ADC, accessed via a Waveshare Raspberry Pi HAT breakout module. The ADC was configured to operate at 7,200 SPS total throughput, cycling across four input channels. With four active channels, the effective per-channel sample rate was approximately 1,800 SPS. For the intended application — monitoring DC signals and low-frequency control waveforms well below 100 Hz — this sample rate exceeds the Nyquist requirement by more than an order of magnitude, and was considered more than sufficient for our use case, with room to improve in the future.

The ADC was configured with a FIR digital filter for hardware-level noise rejection, and the Qt-based application provided a software exponential moving average (EMA) filter, adjustable from 0 to 1.0 in 0.01 steps, allowing students to trade temporal resolution for signal clarity. The ring buffer holds up to 100,000 samples, corresponding to approximately 55 seconds of continuous capture, and

the default display window spans 5 seconds, configurable from 0.1 to 10 seconds. The display refreshes at 25 Hz, which was found to be sufficient for visualizing the slow control signals present in this system.

The primary limitation of the oscilloscope system was residual voltage noise on the displayed waveforms. The ADS1263 accepts a differential input range of ± 2.5 V around its internal 2.5 V reference, with the system's ± 12 V range mapped down through a resistive level-shifting PCB. The additional signal conditioning stage introduced noise that was not fully suppressed by the hardware FIR filter or software EMA, particularly at higher gain settings. The root cause was identified as a combination of shared analog and digital ground routing on the level-shifting PCB and SPI signal integrity issues arising from the relatively long trace paths between the ADC and the Raspberry Pi. A split analog/digital ground plane, shorter SPI traces, and additional decoupling capacitors at the ADC supply pins would likely reduce this noise floor substantially in a future revision.

5.3 Saturation Detection Analysis

Each module (except the switch) incorporates an LM393-based window comparator circuit for saturation detection. The circuit was designed to trigger when the op-amp output approaches within 0.5 V of either supply rail, corresponding to reference voltages of $V_{\text{ref}+} = 11.75$ V and $V_{\text{ref}-} = 0.25$ V relative to the virtual ground reference.

Testing of the saturation detection circuit confirmed that the comparator output remained above approximately 10 V during normal (non-saturated) operation, and fell below 150 mV when saturation was detected, providing a clean digital-like transition suitable for driving the indicator LED and buzzer. The LED turn-on threshold was measured at approximately 7.8 V, and LED forward current at 12 V was measured at 3.4 mA, within the rated limits of both the LED and the LM393 open-collector output. Reference voltage accuracy was confirmed to within ± 25 mV of the target values, and 330 k Ω hysteresis resistors were used to prevent rapid oscillation of the indicator near the saturation boundary. The saturation detection system performed reliably throughout testing and demonstration, and was one of the more polished subsystems in the final product.

5.4 Mechatronic System Analysis

The mechatronic system was validated through physical demonstration of sustained ball levitation. The VL53L1X time-of-flight sensor was configured with a

4-SPAD region of interest (the minimum available) to produce a narrow sensing cone that negated reflections from the interior walls of the acrylic tube. A 50 ms measurement timing budget was used, yielding a sensor update rate of 20 Hz over I^2C at 400 kHz. This update rate was consistent with the control loop bandwidth of the system, as the fan dynamics and ball inertia limited meaningful response to frequencies well below 20 Hz.

The Arduino Nano generated a 25 kHz PWM signal to drive the Noctua NF-F12 fan, a frequency chosen to place the switching noise above the audible range and to match the fan's native PWM specification. Fan duty cycle was mapped linearly from the analog control input, with the PWM compare register ranging from 127 to 255 out of a period of 639 counts (corresponding to roughly 20% to 40% duty cycle), tuned empirically to span the useful levitation range within the tube without the ball bottoming out or escaping the top. The sensor output was converted back to an analog voltage via PWM-filtered DAC on a second output pin and returned to the Analog Computer as the feedback signal, closing the loop.

A known limitation of the mechatronic system is that the Arduino's PWM-based DAC output for the sensor feedback signal introduces quantization and ripple that appears as noise on the feedback path. A true DAC, or an RC filter with a lower cutoff frequency on the PWM output, would improve signal quality. Additionally, the control loop latency introduced by the 50 ms sensor measurement budget, Arduino processing time, and fan spin-up dynamics limited achievable loop bandwidth, which constrained how aggressively the gain could be set before the system became unstable.

5.5 Summary

Overall, the Analog Computer met the requirements that most directly affected the student experience: modular control operations, real-time signal visualization, and a physical demonstration of closed-loop feedback. The subsystems that fell short of their full potential — primarily the oscilloscope noise floor and the DAC resolution on the mechatronic feedback path — were both attributable to PCB-level design decisions and component selection tradeoffs made under time and resource constraints. Both are well-understood problems with known solutions that would be straightforward to address in a future revision.

6 So What?

Logan Caskey

- **Leadership is hard. Really hard.** As the team lead for this project, I really struggled to keep both myself and everyone on track and lead in general. I think I sort of treated the leadership role as somewhat synonymous with a technical lead role at times, which me also having the role of system architect didn't really help that distinction in my mind, and I didn't do enough of the actual leadership side of things at times. It taught me better ways to communicate with my team, and taught me some personal discipline that I don't normally have surrounding these types of projects (in terms of trying to be on top of everything).
- **SPI can be a nightmare** On a more technical note, the biggest technical challenge for me this year by far was working on the oscilloscope. As system architect, I selected just about every single electrical component in our system, on both the computer side and mechatronic side, and I sort of naively chose components at times where I didn't consider how hard it would be to implement, thinking that it would be fine. In particular, the ADC I selected was way overspec'd for our use case, and I wanted the oscilloscope to be overspec'd in general, but I sort of assumed it would be ok and that the integrity of our signal would be fine so long as we do some best practice PCB design things, and that ended up not being the case. Also, in selecting a QFN package component, we were unable to prototype soon, having to order a custom adapter online to be able to prototype on a breadboard, adding a lot of time delay in our project that wasn't ever necessary, instead of choosing a more robust and proven part. As a result, I learned a LOT about SPI communication, its pitfalls, good circuit design practice in high speed digital circuits, the QFN package and how to work with it, and more.
- **There will always be something "not perfect", and things will almost never go the way you plan** As aforementioned with the ADC, there were definitely issues that took place during the duration of our project. I have a bad tendency to be a bit of a perfectionist on certain things, and was getting very frustrated at points through this year when things weren't going the way I wanted or had envisioned them to. In some ways, I think that my perfectionism helped our group to turn out better work at points, but other

times I think it just turned into me having this sort of bad attitude that things weren't turning out the way I wanted, and affected both me and my group negatively. I learned to sort of accept all the flaws that are going to come with our project, and that almost regardless of everything, there was no way I or my group could have lived up to my standards, and that's OK.

Dakota Golden

- **Large projects are much more dependent on teamwork than I originally expected.** Although I had completed group projects before Capstone, this was the first time I worked with the same team on a technically complex project over the course of an entire academic year. One of the biggest takeaways for me was realizing how much can be accomplished when multiple people contribute different skills toward a shared goal. Our final product only came together because responsibilities were distributed across the team, and everyone contributed to solving different technical challenges. Seeing the system progress from an idea into a functioning product reinforced how important communication, collaboration, and shared ownership are in larger engineering projects.
- **PCB assembly and debugging are often harder than the design itself.** A major technical challenge for me involved assembling and troubleshooting PCBs. Designing a board and receiving fabricated hardware felt like a significant milestone, but getting the board to work reliably introduced a completely different set of challenges. Surface-mount soldering required far more consistency and precision than I expected, and when a board failed, it was not always obvious whether the issue came from assembly mistakes, solder quality, or the circuit design itself. This process taught me patience and showed me how important manufacturing quality and structured debugging practices are when working with custom hardware.
- **Subsystem integration is where a project begins to feel real.** One area that stood out to me throughout the project was subsystem integration. Once individual boards and components were verified independently, combining them into a complete system required a different style of problem solving. I became more comfortable tracing signals, diagnosing interactions between subsystems, and validating that separate pieces of hardware worked together as intended. Integration work strengthened my debugging skills because

problems were often caused by the relationship between components rather than a single isolated failure.

- **Planning early saves more time than rushing into implementation.** Early in the project, a significant amount of time was spent developing block diagrams, power budgets, and risk analysis documents. At the time, much of this planning felt slower and less rewarding than immediately building hardware. However, throughout integration and troubleshooting, those early design documents became extremely valuable references. This project taught me that strong planning provides structure for decision making later in development, especially when systems become more complex and multiple subsystems must interact reliably.
- **Testing early is worth the uncertainty.** Looking back, I would have started ordering components and experimenting with hardware much earlier in the design process. Early on, I hesitated to commit to parts without confidence that they would work correctly, but I learned that uncertainty is unavoidable in engineering design. Earlier testing would have allowed us to identify risks sooner and iterate more effectively. This experience reinforced that prototyping and experimentation are not separate from design; they are essential parts of it.

Joel Ramos Silva

- **Mechanical design has to work closely with the electrical system.** One of the biggest things I learned from this project was that mechanical design cannot happen separately from the rest of the system. Throughout development, enclosure size, mounting locations, connector placement, and internal spacing all changed as the electrical design evolved. Small mechanical decisions often affected how easy the system was to assemble and use, which showed me how important communication is between different parts of the project.
- **Physical assembly reveals problems that CAD does not always show.** Designing parts in CAD made it easy to visualize how everything should fit together, but many challenges only became clear once real hardware was assembled. Tolerances, alignment issues, wire routing, and spacing problems often appeared during integration even when the design looked correct on screen. This taught me that prototyping and test fitting are essential parts

of mechanical design, especially when multiple subsystems must fit into a limited space.

- **Manufacturing and assembly matter just as much as functionality.** A major takeaway from this project was learning to design parts that are not only functional, but also practical to build and assemble. Some components worked well in theory but became difficult to print, mount, or repeatedly install during testing. Over time, I learned to think more about assembly order, hardware access, material limitations, and long-term durability when creating mechanical parts.
- **Iteration is a normal part of the design process.** Very few mechanical components worked perfectly on the first attempt. Multiple revisions were needed to improve fit, strength, and overall usability. While redesigning parts could sometimes feel repetitive, this project taught me that iteration is a normal part of engineering and often leads to a much stronger final result.

Eric Chen

- **Strong teamwork is what allows a large project to succeed.** One of the biggest takeaways from this project was learning how much can be accomplished when a team works well together over a long period of time. Developing the Analog Computer required consistent collaboration between multiple subsystems, and progress depended on everyone contributing to a shared goal. Working alongside the same group over the course of the year showed me that a coordinated team can solve problems and build a much stronger final product than any individual could accomplish alone.
- **Planning early creates flexibility later in the project.** Throughout the design process, I learned that planning before implementation is extremely important. Early decisions surrounding timelines, component selection, and subsystem dependencies had a large impact later during integration. Having clear plans and backup options helped reduce uncertainty when challenges appeared. This project reinforced that time spent organizing and preparing is not lost time; it creates opportunities to troubleshoot and adapt when things do not work as expected.
- **PCB assembly takes more time and practice than expected.** One of the more difficult technical areas for me involved PCB design and soldering.

While designing and ordering boards felt straightforward, assembling and debugging them required significantly more time than anticipated. Small soldering issues or component placement errors could create problems that were difficult to diagnose. This taught me that manufacturing and assembly skills are just as important as circuit design, especially when deadlines are involved.

- **Asking for help is part of good engineering practice.** A personal lesson I gained from this project was becoming more comfortable asking questions when I was unsure about something. Early on, I often tried to solve problems independently for too long, which sometimes slowed progress. Over time, I realized that reaching out to teammates or seeking advice usually led to faster solutions and stronger collaboration. Learning to ask for help improved both my efficiency and my confidence when working within a technical team.

Zanghyeoun Shim

- **PCB design requires mechanical awareness, not just electrical correctness.** As the PCB Design Manager, I quickly learned that a schematic being electrically correct is only the starting point. Physical constraints — pad dimensions, board outline, component placement relative to the enclosure, and pogo-pin contact geometry — all had to be considered simultaneously. Designing a PCB that fits and functions inside a real mechanical housing was a genuinely different challenge from anything covered in coursework, and it made the work much more interesting.
- **Hands-on testing and debugging are how system reliability is actually built.** Iterating through test-fail-debug cycles on the fabricated boards taught me that reliability is not something you design in once and forget — it is something you verify repeatedly. Each debugging session deepened my understanding of where real hardware diverges from simulation, and gave me practical intuition that I could not have gained any other way.
- **Noise reduction is a first-class design concern in mixed-signal PCBs.** The most technically demanding part of my work was the ADC PCB for the integrated oscilloscope. Achieving clean waveforms on the LCD screen required minimizing noise at the ADC input, which meant the board layout itself had to do real engineering work. I created the ADC schematic symbol

and PCB footprint library from scratch, studied multiple datasheets, and sought expert advice before settling on a split analog/digital ground plane strategy. Seeing the improvement in signal quality after implementing the ground separation was one of the most satisfying moments of the project.

- **Owning the full PCB lifecycle is an invaluable experience.** Taking a design from initial schematic capture all the way through layout, fabrication order, assembly, bring-up testing, debugging, and final integration into the complete system gave me a complete picture of what professional PCB development looks like end-to-end. There is still a great deal more to learn as a PCB designer, but having navigated the entire process once — with real stakes — is a foundation I will build on throughout my career.

7 Journey Reflection

Throughout the development of the Analog Computer, our team learned that large engineering projects depend just as much on communication and coordination as they do on technical ability. Because the system combined analog electronics, PCB design, embedded software, mechanical integration, and a mechatronic demonstration, progress relied heavily on collaboration between team members.

One process that worked particularly well was dividing ownership of subsystems while maintaining regular communication. Each team member had a primary area of responsibility, allowing work to progress in parallel while remaining connected to the larger system. Weekly meetings and troubleshooting sessions helped keep development aligned with project goals.

Brainstorming was most effective early in the design process when the project scope was still flexible. Ideas were evaluated on technical feasibility, educational value, manufacturability, and usability. Rather than committing immediately to a single approach, the team benefited from comparing multiple options before implementation.

Decision making followed a balance between technical ownership and group consensus. Subsystem owners often led detailed design decisions, while larger architectural choices were discussed collectively. This allowed expertise to guide implementation without slowing progress through excessive debate.

One of the largest lessons learned involved the importance of testing and iteration. Many assumptions about integration, signal quality, and manufacturability changed once physical hardware existed. The project reinforced that debugging is not a separate stage after design, but a continuous process that shapes the final system.

If completing a similar project again, the team would begin prototyping and subsystem validation earlier. Earlier testing would have reduced uncertainty and created more time for refinement rather than late-stage troubleshooting. Stronger documentation between subsystems would also simplify integration and debugging.

Overall, the project demonstrated that successful engineering comes from planning, communication, experimentation, and persistence. While the Analog Computer met its primary goals, the larger takeaway was learning how to organize, adapt, and solve problems together while developing a complex system from concept to implementation.